Data Engineering [D.E]

Assigment - 02

Name: A. Aswini
Branch: CSD
Roll No: 25NUSA4401

1. Discuss briefly about generation phase and storage phase in data engineering life cycle.

Ans: Generation:

- A source is the origin of the data used in the data engineering life cycle. A source system can be an transactional data base or IOT device etc.
- Engineers talk to app owners to ensure code changes don't break the data flow.
- Engineers must work with the people who build them, so the data flow never stops or breaks.

Storage:

- It connects every part of the life cycle (Ingestion, transformation & serving).
- Data is stored by "temperature", where "Hot" data is kept in fast storage for daily use & "cold" data is kept in cheap storage for archives, as no single system fits every speed or cost need.
- Cloud storage choice - Pick a storage solution (like data warehouse, data lake) that matches your needs for speed, scale and easy access in the cloud.

2) Explain the role of data transformation & serving data in the data engineering life cycle.

Ans: Transformation:

It is the process of changing raw data from its original form into a clean, useful format for reports, analysis or ML

(1) 1

- Cleaning: Fixing dates and numbers to read data from its original form into a clean report
- Value & ML: Applying business logic & preparing features for AI/ML models.

Serving:
Data only has value when it is actually used for practical work; storing massive amounts of data without a plan to use it a "vanity project" that wastes money.
Most popular uses of data are: Analytics, ML & Reverse ETL.

3) Explain software engineering and orchestration as underpinnings in data engineering life cycle?

A: Software engineering:
Software engineering provides technical foundation for building data pipelines and data platforms. DE systems are developed using programming languages and frameworks and therefore require proper software development practices.
Software engineering ensures modular code design, version control, testing, documentation and error handling across all stages of D.E life cycle.

a) Streaming: Streaming means processing data in real time instead of waiting to collect it in batches.
It uses special tools & methods to handle continuous data flow.

b) Pipeline as a code: Data engineers write code to define and manage data tasks automatically instead of them doing them manually.

General purpose problem solving:

Data engineers ~~often~~ face special cases not covered by tools, so they must write custom code.

Orchestration:

Orchestration serves as the vital co-ordination layer, managing complex workflows by tracking job dependencies through directed acyclic graphs.

Orchestration handles task dependencies, automation, retries and failure recovery, reducing manual intervention. It ensures consistent dataflow, timely availability of data and operational efficiency across all stages of the life cycle